THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The converse of the proposition "If a number is divisible by 10, then it is divisible by 5." is: 1

(a) If a number is divisible by 5, then it is divisible by 10.
(b) If a number is not divisible by 5, then it is not divisible by 10.
(c) If a number is not divisible by 10, then it is not divisible by 5.
(d) A number is divisible by 10 only if it is divisible by 5.

2. The converse of the proposition "If a number is a perfect square, then it is non-negative." is: 1

(a) Every non-negative number is a perfect square and vice versa.
(b) If a number is non-negative, then it is a perfect square.
(c) If a number is negative, then it is not a perfect square.
(d) If a number is not a perfect square, then it is negative.

3. Which of the following is a counterexample to the proposition "If n is a prime number, then $n + 2$ is also a prime number."? 1

(a) 7 **(b)** 11 **(c)** 3 **(d)** 5

4. The converse of "If n is even, then n^2 is even" is "If n^2 is even, then n is even". Which statement is correct? 1

(a) both the proposition and its converse are true
(b) only the converse is true
(c) only the proposition is true
(d) both are false

5. A triangle has sides 20 cm, 21 cm and 29 cm. By the converse of the Baudhāyana–Pythagoras theorem it is: 1

(a) not right-angled, since the sides are not equal
(b) right-angled, since $a + b > c$
(c) right-angled, since all sides are different
(d) right-angled, since $a^2 + b^2 = c^2$

6. The number 49 has: 1

(a) 5 factors; it is a perfect square **(b)** 4 factors; it is not a perfect square
(c) 3 factors; it is a perfect square **(d)** 5 factors; it is not a perfect square

7. The number of factors of 121 is: 1

(a) 2 **(b)** 3 **(c)** 11 **(d)** 4

8. The statements "If X then Y" and "X implies Y" have: 1

(a) unrelated meanings **(b)** opposite meanings
(c) the same meaning **(d)** the meaning of the converse

9. It is possible that a proposition and its converse are: 1

(a) both true, or both false, or one true and one false
(b) only both true
(c) only one true
(d) never both false

10. Which of the following converses is true? 1

(a) If two triangles have equal areas, they are congruent.

(b) If a number is divisible by 3, it is divisible by 6.

(c) If the angles opposite two sides of a triangle are equal, the two sides are equal.

(d) If a number is prime, it is odd.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the converse of "If two lines are parallel, then the alternate interior angles formed by a transversal are equal." Is the converse true? 2

12. State the converse of "If a number is divisible by 18, then it is divisible by 9." Is it true? Justify. 2

13. Write the converse of "If $x = y$, then $x^2 = y^2$" and give a counterexample for it. 2

14. Give a counterexample to show that "All numbers of the form $n^2 + n + 11$ are prime" is false. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If a number is divisible by 15, then it is divisible by 5." Decide whether each is true and give a counterexample if one is false. 3

16. Write the converse of "If two triangles are congruent, then their corresponding angles are equal." Is the converse true? Give a counterexample if false. 3

17. Show that the number 143 gives a counterexample to the claim "$n^2 + n + 11$ is prime for every natural number n". State the value of n used. 3

OR

Test whether the triangle with sides 12 cm, 35 cm, 37 cm is right-angled. Name the theorem you use. 3

18. Write the converse of "If a number is a multiple of 6, then it is a multiple of 3". Justify the proposition and give a counterexample for the converse. 3

OR

List the factors of 36 as factor–partner pairs. Which pair repeats a number? What does this tell you about the number of factors? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Anu writes: "If it rains, the road is wet." 5

(i) Write the converse.

(ii) Give a situation showing the converse can be false.

(iii) Is the original proposition always true?

(iv) Define a counterexample and explain how it helps.

OR

Prove that the converse of the Baudhāyana–Pythagoras theorem is true: if a triangle ABC has $a^2 + b^2 = c^2$, then it is right-angled. 5

20. (i) Show that a perfect square has an odd number of factors by pairing factors, using the number 49 and the number 100. 5

(ii) Write the converse and say whether it is true.

OR

Case study: Ravi tests the formula $n^2 + n + 11$ for $n = 1, 2, 3, 4$ and finds 13, 17, 23, 31, all prime. 5

(i) Compute these values.

(ii) Can Ravi conclude the formula always gives primes?

(iii) Find the smallest n in $\{1, \ldots, 11\}$ for which the value is not prime.

(iv) Name the idea of the example you found.

END OF QUESTION PAPER — Total: 40 Marks